

Athiyaman M

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Professional Summary

Software Engineer with experience spanning healthcare technology, clinical software development, artificial intelligence, and database engineering. Currently contributing to national-scale software infrastructure projects at C-DAC while previously developing healthcare applications at HTIC, IIT Madras. Interested in AI-driven healthcare systems, medical imaging, clinical decision support, and data-centric healthcare research.

Education

B.E. ECE (Honours in AI/ML)

IFET College of Engineering, Anna University

11/2021 – 05/2025 | Tamil Nadu, India

CGPA : **8.71** | First Class with Distinction | Rank Holder | Associate Secretary

Professional Experience

Project Engineer (Software)

C-DAC, Government of India

08/2025 – Present | Chennai, India.

- Developed backend Python Script for the SSC Exam Conduction Project mainly focused on security flags checks, score generation and blocking anonymous clients by reading, writing and generating logs. Developed proposal projects such as AuditShakti, Research Paper Summarization, Mental Health Prediction with XAI as ready POC.
- Contributing to BharatDB project, a government initiative to build database for India by extending and optimizing PostgreSQL open-source codebase. Resolved open issues and delivered patches in C and SQL, improving database functionality and stability in alignment with community standards of PostgreSQL.

Project Intern (Software)

HTIC, IIT Madras

06/2024 – 07/2025 | Chennai, India.

- Developing an desktop application for TKA project using QT framework with VTK/ITK library in C++ and managing MySQL database and Azure DevOps for CI/CD pipeline and testing our software. Utilizing QT Creator for debugging, Visual Studio, CMake for configuration, C++ and QML languages.

Projects

[1. Mental Health Risk Detection via Explainable AI \(XAI\)](#)

- Developed an NLP-based mental health risk detection system using BERT embeddings and machine learning models to identify at-risk individuals from social media text. Achieved 87% AUC-ROC while incorporating SHAP, LIME, and Counterfactual Analysis for interpretable and trustworthy predictions.
- **Tech Stack** : Python, BERT, Scikit-Learn, SHAP, LIME, Pandas, Streamlit

[2. RAG-Based LLM Document Question Answering System](#)

- Developed a document intelligence platform that extracts information from PDFs and generates contextual answers using semantic retrieval and large language models. Implemented vector search for accurate knowledge retrieval from unstructured documents.
- **Tech Stack** : Python, LangChain, Pinecone, Ollama, Llama 2, FAISS

[3. AI-Powered Research Paper Intelligence System](#)

- Developed an intelligent research assistant that summarizes scientific papers, extracts key insights, and enables contextual question answering using Retrieval-Augmented Generation (RAG). Improved literature review efficiency through semantic search and document intelligence.
- **Tech Stack** : Python, LangChain, Ollama, Pinecone, Streamlit, LLMs

[4. ReAct based Personal Agentic AI Query System.](#)

- Built an autonomous AI assistant capable of multi-step reasoning, web search, and dynamic question answering using the ReAct framework. Integrated APIs and external tools to deliver production-ready AI workflows.
- **Tech Stack** : Python, LangGraph, FastAPI, Streamlit, OpenAI, Groq, Tavily

[5. AuditShakti : AI-Powered Audit Intelligence Platform](#)

- Designed an AI-driven compliance and audit assistant to automate document review, risk assessment, and audit report generation. Developed proof-of-concept workflows to improve efficiency in regulatory and compliance analysis.
- **Tech Stack** : Python, LangChain, Streamlit, Vector Databases, LLMs

Research Work

1. Mental Health Risk Detection via Explainable AI (XAI): A Multi-Model Benchmark with Contextual Embeddings

International Conference on Artificial Intelligence Trends and Pattern Recognition (ICAITPR 2026) - Manuscript Under Review

Developed an NLP-based mental health risk detection system using BERT embeddings and machine learning models, achieving 87% AUC-ROC. Integrated SHAP, LIME, and Counterfactual Analysis to provide interpretable and trustworthy predictions for AI-assisted mental health assessment.

2. AI-Based Analysis of Hospital Bed Utilization Patterns for Patient Readmission Prediction

International Conference on Smart AI and Health Innovation Through Tamil for Holistic Advancement (SAHITHYA 2025), NITTTTR Chennai (ISBN Conference Proceedings) - Proceedings (Accepted)

Investigated the application of machine learning models for predicting hospital readmissions using patient occupancy and healthcare operational data. The optimized Gradient Boosting model achieved 94.1% accuracy and provided actionable insights for real-time risk assessment, healthcare resource optimization, and improved clinical decision-making.

3. Design of Passive Optical Network (PON) for Hospital Building Management: Enhancing Data Transmission Using Free Space Optics (FSO)

International Conference on Human-Centric Smart Computing (ICHCSC 2025), Springer Nature - Proceedings (Accepted)

Developed and evaluated a hybrid PON-FSO communication architecture for smart hospital infrastructures, enabling reliable, low-latency, and scalable data transmission for healthcare applications. Analyzed network performance using BER, SNR, and Quality Factor metrics under varying environmental conditions to support next-generation hospital communication systems.

Achievements & Awards

- Awarded the **Rank Holder Award** and **Academic Excellence Award** for outstanding academic performance.
- Recognized as the **Best Student** for consistent academic excellence, leadership contributions, and overall achievements.
- Received the **Best Innovation Project Award** at Project Expo'25 for developing an impactful technology-based solution.
- Awarded **Best Presentation** in the Technical Talk Competition for excellence in technical and research presentation.
- Secured **Runner-Up** position in the CEMP Project Expo 2022 for innovative project development and implementation.
- **Finalist** in a National-Level Mathematics Competition, demonstrating strong analytical and problem-solving abilities.
- Earned the LinkedIn **Top Voice Badge** on three occasions for sharing technical knowledge and industry insights.
- **Recognized** by C-DAC for outstanding work performance and contributions to software development projects.